

mb-pc40s-2019-jun

Typst conversion review. OCR content and mathematical accuracy require editorial review.

Question 1

Avery has 4 adventure books, 5 mystery books, and 1 comic book.

Determine the number of ways he can arrange all of the books on his shelf if each type of book must be grouped together.

Solution

3!	4!	• 5!	• 1!	= 17280 ways
types of books	adventure books	mystery books	comic book	

1 mark for arrangement of types of books

1 mark for arrangement of adventure,
mystery, and comic books

2 marks

Note:

- 1! does not need to be shown.

Question 2

Solve the following equation, algebraically, over the interval $[0, 2\pi]$.

$$5 \cos^2 \theta - \cos \theta - \sin^2 \theta = 0$$

Solution

Use $\sin^2 \theta = 1 - \cos^2 \theta$:

$$5 \cos^2 \theta - \cos \theta - (1 - \cos^2 \theta) = 0$$

$$6 \cos^2 \theta - \cos \theta - 1 = 0$$

$$(3 \cos \theta + 1)(2 \cos \theta - 1) = 0$$

Hence $\cos \theta = -\frac{1}{3}$ or $\cos \theta = \frac{1}{2}$.

For $\cos \theta = -\frac{1}{3}$, the solutions on $[0, 2\pi]$ are

$$\theta = \arccos\left(-\frac{1}{3}\right) \approx 1.911, \quad \theta = 2\pi - \arccos\left(-\frac{1}{3}\right) \approx 4.373.$$

For $\cos \theta = \frac{1}{2}$, $\theta = \frac{\pi}{3}$ or $\theta = 5\frac{\pi}{3}$.

Thus $\theta \approx 1.047, 1.911, 4.373, 5.236$ radians.

Scoring: 1 mark for the identity substitution; 1 mark for solving for $\cos \theta$; 2 marks for the four angle values.

Question 3

Given that $-6048x^2y^5$ is the sixth term in the expansion of $(3x - m)^7$, determine m .

Solution

$$-6048x^2y^5 = {}_7C_5(3x)^2(-m)^5 \quad 2 \text{ marks (1 mark for } {}_7C_5; 1/2 \text{ mark for each consistent factor)}$$

$$-6048x^2y^5 = 21(9x^2)(-m^5)$$

$$-6048x^2y^5 = -189x^2m^5$$

$$32y^5 = m^5 \quad 1/2 \text{ mark for simplification}$$

$$2y = m \quad 1/2 \text{ mark for } m$$

3 marks

Question 4

A series of blood tests measures the concentration of a prescribed drug. This concentration decreases according to the formula $A = Pe^{rt}$ where:

A is the concentration at time t

P is the initial concentration

r is the rate of change

t is the time, in hours, after the first blood test

The initial concentration is 3.8900 units /mL. Three hours later, the concentration is 1.7505 units /mL.

a) Determine the rate of change, r , algebraically.

b) Determine the concentration of the prescribed drug four hours after the initial concentration of 3.8900 units /mL was measured. Express the answer correct to 4 decimal places.

Note: A calculator is not required for the remaining test questions.

Solution

a)

$$1.7505 = 3.8900e^{r(3)} \quad 1/2 \text{ mark for substitution}$$

$$0.45 = e^{3r}$$

$$\ln 0.45 = 3r \ln e \quad 1/2 \text{ mark for applying logarithms}$$

$$\frac{\ln 0.45}{3} = r \quad 1/2 \text{ mark for power law}$$

$$1/2 \text{ mark for value of } r$$

or

$$-0.266169... = r$$

2 marks

b)

$$A = 3.8900e^{-0.266169(4)}$$

$$A = 1.341424...$$

$$A = 1.3414 \text{ units /mL}$$

1 mark for answer consistent with a)

Question 5

Ariane uses the formula $s = \theta r$ to determine the arc length of a circle that has a central angle of 20° and a radius of 15 cm .

Below is Ariane's work:

$$S = \theta r$$

$$S = (20)(15)$$

$$S = 300 \text{ cm}$$

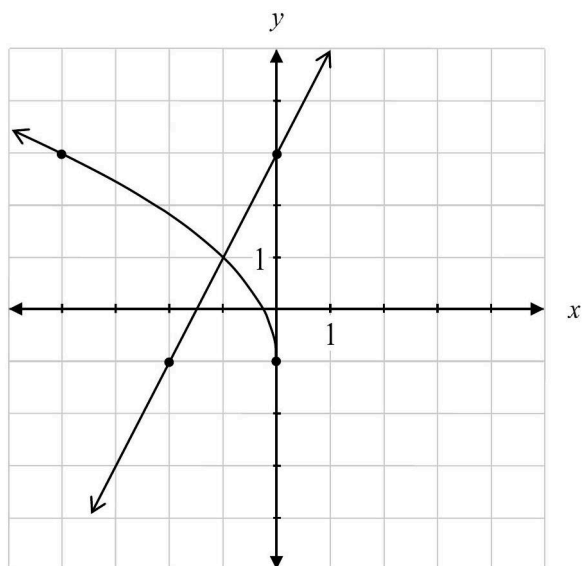
Describe her error.

Solution

When using the formula $s = \theta r$, the angle must be in radians. Ariane did not convert the central angle from degrees to radians.

Question 6

Using the graphs below, state the solution of the equation $2x + 3 = 2\sqrt{-x} - 1$.



Solution

$$x = -1$$

1 mark

Question 7

Solve, algebraically.

$$\log_2(x+3) = 5 - \log_2(x-1)$$

Solution

Method 1

$$\log_2(x+3) + \log_2(x-1) = 5$$

$$\log_2[(x+3)(x-1)] = 5$$

$$x^2 + 2x - 3 = 2^5$$

$$x^2 + 2x - 3 = 32$$

$$x^2 + 2x - 35 = 0$$

$$(x+7)(x-5) = 0$$

$\frac{1}{2}$ mark for solving for the permissible value of x

$\frac{1}{2}$ mark for showing the rejection of the extraneous root

3 marks

Method 2

$$\log_2(x+3) + \log_2(x-1) = 5$$

$$\log_2[(x+3)(x-1)] = \log_2 2^5$$

$$x^2 + 2x - 3 = 32$$

$$x^2 + 2x - 35 = 0$$

$$(x+7)(x-5) = 0$$

$$x = 5$$

$\frac{1}{2}$ mark for the permissible value of x

$\frac{1}{2}$ mark for showing the rejection of the extraneous root

3 marks

Question 8

Explain why the value of n must be greater than or equal to the value of r , when using C_n .

Solution

The number of objects to select, r , cannot be greater than the total number of objects, n .

1 mark

Question 9

3 marks

110

Given that $y = |x|$, determine the equation of the resulting function, $g(x)$, after the following transformations:

- reflection in the x -axis
- vertical translation 5 units down
- horizontal stretch by a factor of 3

$g(x) =$

Solution

$$g(x) = -\left|\frac{1}{3}x\right| - 5$$

1 mark for vertical reflection

1 mark for vertical translation

1 mark for horizontal stretch

3 marks

Question 10

Explain why the graph of $y = \frac{x-1}{x^2+x-6}$ has a horizontal asymptote at $y = 0$.

Solution

As x approaches positive or negative infinity, y approaches zero.

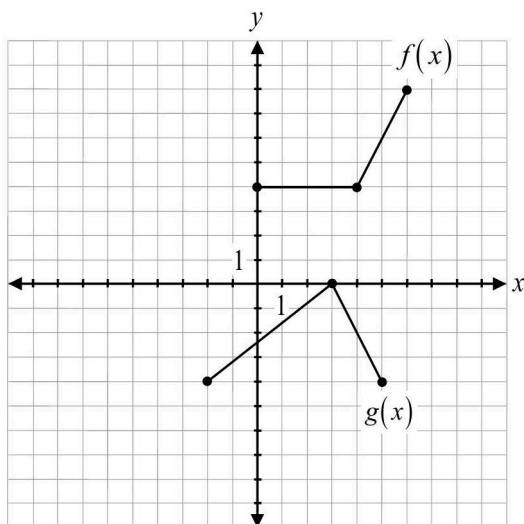
1 mark

or

The degree of the numerator is less than the degree of the denominator.

Question 11

Given the graphs of $f(x)$ and $g(x)$, evaluate $g(f(2))$.



Solution

$$f(2) = 2 \quad \frac{1}{2} \text{ mark for the value of } f(2)$$

$$g(2) = -1 \quad \frac{1}{2} \text{ mark for consistent value of } g(f(2))$$

1 mark

Question 12

Kennedy was asked to solve the equation $\tan \theta = 1$ over all real numbers.

Below is Kennedy's solution:

$$\tan \theta = 1$$

$$\theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

Describe her error.

Solution

Kennedy did not include the general solution in her answer.

Question 13

Solve, algebraically.

${}_nC_3 = 3({}_nP_2)$

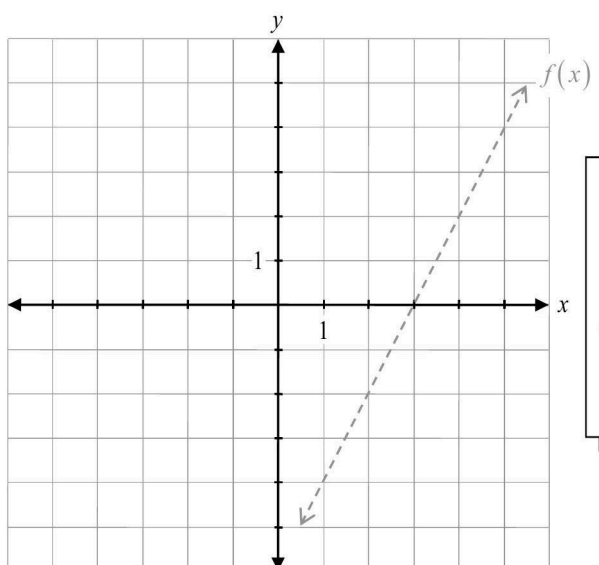
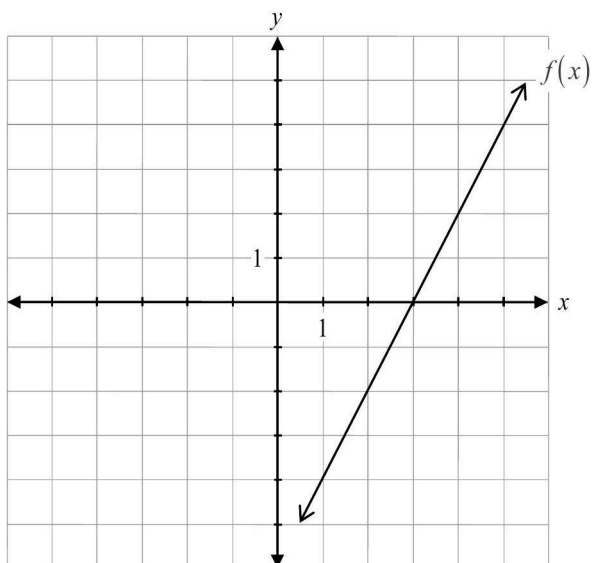
Solution

$\frac{n!}{(n-3)!3!} = \frac{3n!}{(n-2)!}$	1 mark for substitution into equation ($\frac{1}{2}$ mark for each side)
$\frac{n!(n-2)!}{(n-3)!3!} = 3n!$	
$\frac{n!(n-2)(n-3)!}{(n-3)!} = 3!(3n!)$	1 mark for factorial expansion
$n - 2 = 3!(3)$	$\frac{1}{2}$ mark for simplification of factorials
$n - 2 = 18$	
$n = 20$	$\frac{1}{2}$ mark for solving for n

3 marks

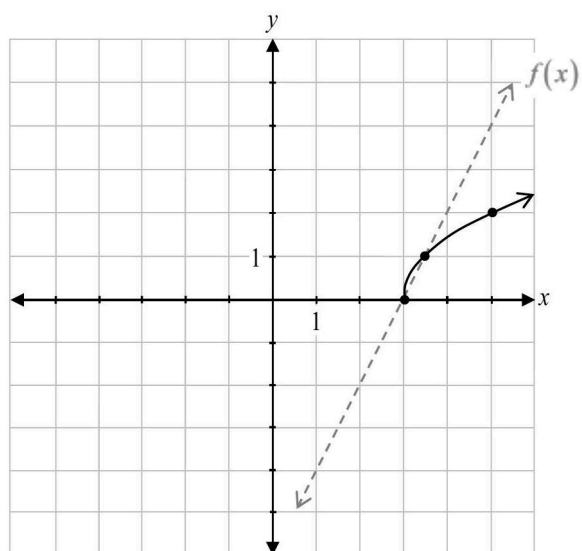
Question 14

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

Solution



1 mark for restricting domain

$\frac{1}{2}$ mark for shape between invariant points

$\frac{1}{2}$ mark for shape to the right of invariant points

2 marks

Question 15

Prove the identity for all permissible values of θ .

$$\frac{\sec \theta - \tan \theta \sin \theta}{\tan \theta \sin \theta} = \csc^2 \theta - 1$$

Left-Hand Side	Right-Hand Side

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{\frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta} \cdot \sin \theta}{\frac{\sin \theta}{\cos \theta} \cdot \sin \theta}$ $\frac{\frac{1 - \sin^2 \theta}{\cos \theta}}{\frac{\sin^2 \theta}{\cos \theta}}$ $\frac{1 - \sin^2 \theta}{\sin^2 \theta}$ $\frac{1}{\sin^2 \theta} - \frac{\sin^2 \theta}{\sin^2 \theta}$ $\csc^2 \theta - 1$	$\csc^2 \theta - 1$

1 mark for substitution of appropriate identities

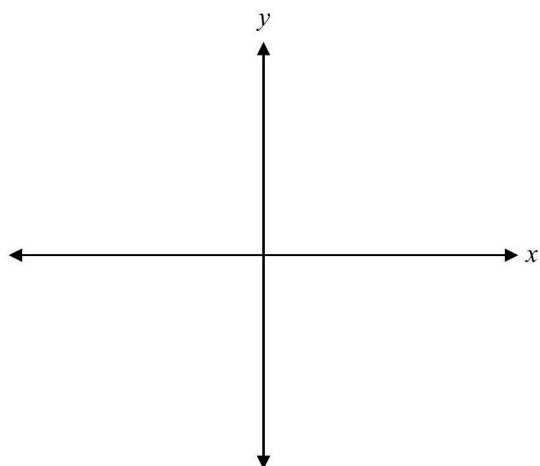
1 mark for algebraic strategies

1 mark for logical process to prove the identity

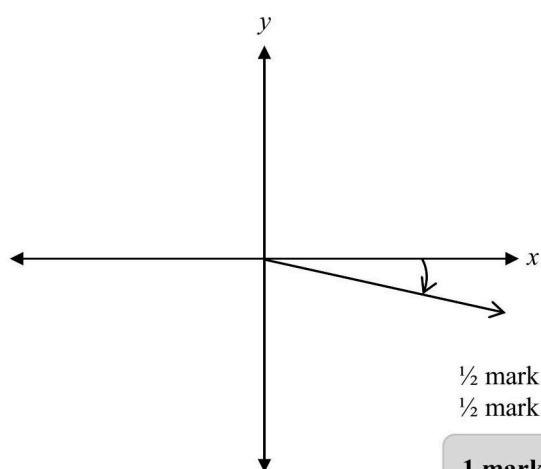
3 marks

Question 16

Sketch the angle of $-\frac{\pi}{12}$ radians in standard position.



Solution



$\frac{1}{2}$ mark for an appropriate angle in quadrant IV
 $\frac{1}{2}$ mark for correct direction

1 mark

Note:

- If the directional arrow is not indicated, deduct an E1 error (final answer not stated)

Question 17

Given that $h(x) = 2x^2 - 7x - 15$, determine possible equations of the functions $f(x)$ and $g(x)$ if $h(x) = f(x) \cdot g(x)$.

$f(x) =$

$g(x) =$

Solution

$f(x) = \underline{2x + 3}$ 1 mark for two correct factors of $h(x)$

$$g(x) = x - 5$$

1 mark

Note:

- Other answers are possible.

Question 18

The range of $y = f(x)$ is $-6 \leq y \leq 12$. The range of the transformed function $y = af(x)$ is $-2 \leq y \leq 4$. Identify the value of a .

A.

-3

B.

$-\frac{1}{3}$

C.

$\frac{1}{3}$

D.

3

Solution

Correct answer: C

$\frac{1}{3}$

Question 19

Identify the expression which is equivalent to $3 \log y - \frac{1}{2} \log x + \log z$.

A.

$$\log\left(\frac{y^3}{\sqrt{xz}}\right)$$

B.

$$\log\left(\frac{y^3 z}{\sqrt{x}}\right)$$

C.

$$\log\left(\frac{y^3}{x^2 z}\right)$$

D.

$$\log\left(\frac{y^3 z}{x^2}\right)$$

Solution

Correct answer: B

$$\log\left(\frac{y^3 z}{\sqrt{x}}\right)$$

Question 20

Identify the measure of the angle $-\frac{2\pi}{9}$ in degrees.

A.

$$-400^\circ$$

B.

$$-40^\circ$$

C.

$$40^\circ$$

D.

$$320^\circ$$

Solution

Correct answer: B

$$-40^\circ$$

Question 21

If $y = f(x)$ has a domain of $[2,5]$ and a range of $[6,10]$, identify the domain of $y = f^{-1}(x)$.

A.

$$\left[\frac{1}{2}, \frac{1}{5}\right]$$

B.

$$[-5, -2]$$

C.

$$[-10, -6]$$

D.

$$[6, 10]$$

Solution

Correct answer: D

$$[6, 10]$$

Question 22

Identify which of the following is a polynomial function.

A.

$$p(x) = -\frac{1}{2}(x+2)^3(x-3)$$

B.

$$p(x) = 2x^{\frac{1}{2}} + x - 3$$

C.

$$p(x) = 3x^{-4} + x^2 - 6$$

D.

$$p(x) = 2^x + 3$$

Solution

Correct answer: A

$$p(x) = -\frac{1}{2}(x+2)^3(x-3)$$

Question 23

Identify the total number of terms in the expansion of $(x - y)^9$.

A.

8

B.

9

C.

10

D.

11

Solution

Correct answer: C

10

Question 24

Identify the exact value of $2 \cos^2(15^\circ) - 1$.

A.

$$1$$

B.

$$\frac{1}{2}$$

C.

$$\frac{\sqrt{3}}{2}$$

D.

$$\sqrt{3}$$

Solution

Correct answer: C

$$\frac{\sqrt{3}}{2}$$

Question 25

The zeros of the function $y = f(x)$ are $x = -2$ and $x = 3$. Identify the zeros of the function

$$g(x) = 2f(x - 4).$$

A.

$$x = -6 \text{ and } x = -1$$

B.

$$x = 2 \text{ and } x = 7$$

C.

$$x = -4 \text{ and } x = 6$$

D.

$$x = 0 \text{ and } x = 10$$

Solution

Correct answer: B

$$x = 2 \text{ and } x = 7$$

Question 26

Identify the value of $\log_4\left(\frac{1}{4}\right)$.

A.

−16

B.

−1

C.

1

D.

16

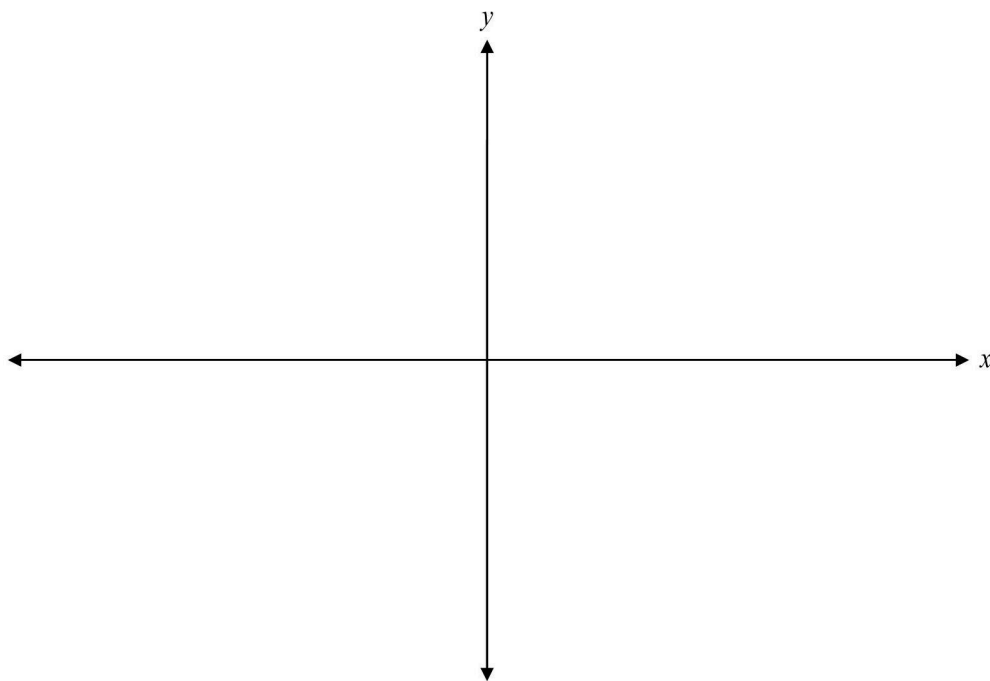
Solution

Correct answer: B

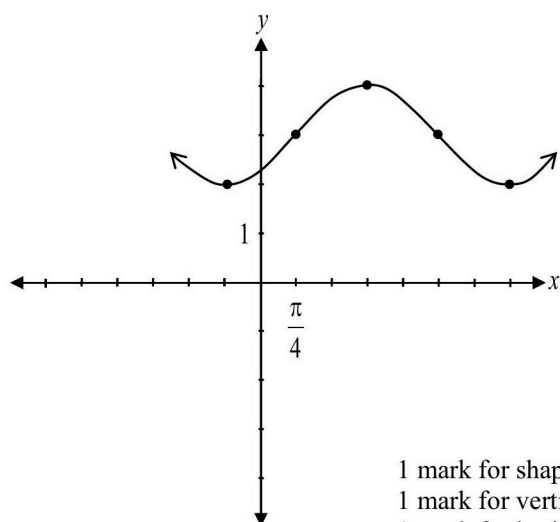
−1

Question 27

Sketch the graph of at least one period of the function $y = -\cos\left(x + \frac{\pi}{4}\right) + 3$.



Solution



- 1 mark for shape of a sinusoidal function with correct period
- 1 mark for vertical reflection
- 1 mark for horizontal translation
- 1 mark for vertical translation

4 marks

Question 28

Justify that $(x - 5)$ is not a possible factor of the function $P(x) = x^3 - 3x^2 - 4x + 12$.

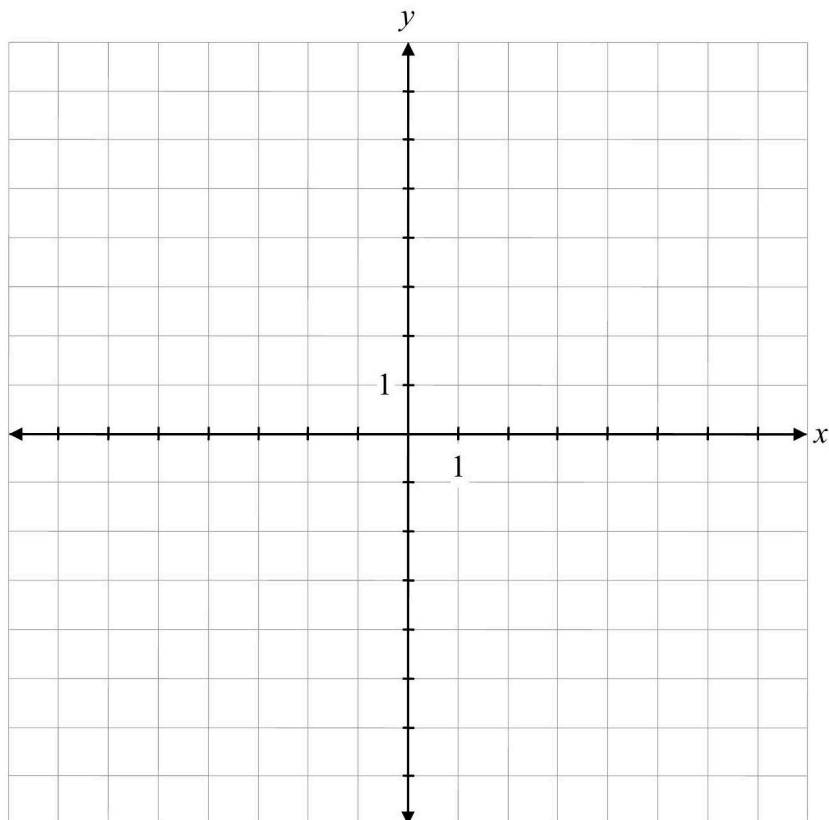
Solution

When $x = 5$ is substituted into $P(x)$, $P(5)$ does not equal 0 .

1 mark

Question 29

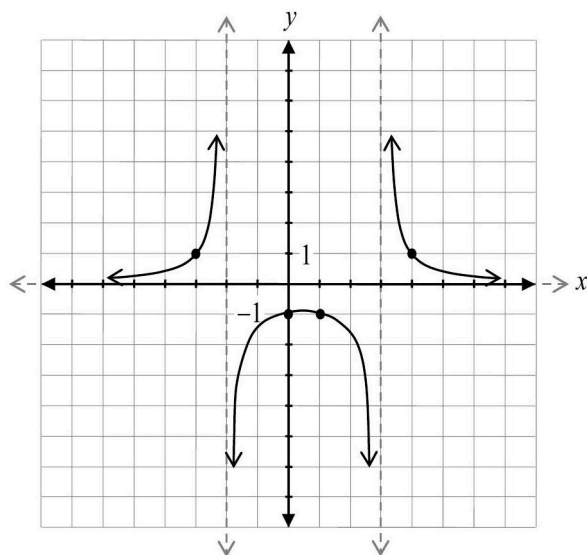
Sketch the graph of $f(x) = \frac{6}{(x+2)(x-3)}$ and state the y -intercept.



y -intercept:

—

Solution



—

-1

1 mark for vertical asymptotic behaviour

($\frac{1}{2}$ mark for behaviour approaching $x = -2$; $\frac{1}{2}$ mark for behaviour approaching $x = 3$)

1 mark for horizontal asymptotic behaviour approaching $y = 0$

$1\frac{1}{2}$ marks for shape ($\frac{1}{2}$ mark for shape in each section)

$\frac{1}{2}$ mark for y -intercept

4 marks

Question 30

Determine how many 3 -digit odd numbers less than 300 are possible using the digits 1, 2, 3, 4, 5, 6
if repetition is not allowed.

Solution

$$\text{case 1 : } \frac{1}{1} \bullet \frac{4}{1} \bullet \frac{2}{\text{odd}} = 8$$

$\frac{1}{2}$ mark for case 1

$$\text{case 2 : } \frac{1}{2} \bullet \frac{4}{1} \bullet \frac{3}{\text{odd}} = 12$$

$\frac{1}{2}$ mark for case 2

$$8 + 12 = 20 \text{ numbers}$$

1 mark for addition of cases

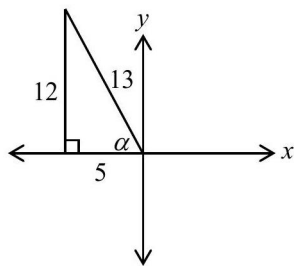
2 marks

Question 31

Given that $\cos \alpha = -\frac{5}{13}$ and $\sin \beta = \frac{2}{3}$, where α and β terminate in the same quadrant, determine the exact value of $\cos(\alpha - \beta)$.

determine the exact value of $\cos(\alpha - \beta)$.

Solution



$$\begin{array}{c} y \\ \uparrow \\ \frac{\sqrt{5}}{3} \\ \downarrow \\ x \end{array}$$

$$x^2 + y^2 = r^2$$

$$25 + y^2 = 169$$

$$y^2 = 144$$

$$y = \pm 12$$

$$x^2 + y^2 = r^2$$

$$x^2 + 4 = 9$$

$$x^2 = 5$$

$\frac{1}{2}$ mark for value of y

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$= \left(-\frac{5}{13}\right) \left(-\frac{\sqrt{5}}{3}\right) + \left(\frac{12}{13}\right) \left(\frac{2}{3}\right)$$

$$= \frac{5\sqrt{5}}{39} + \frac{24}{39}$$

$$= \frac{5\sqrt{5} + 24}{39}$$

$\frac{1}{2}$ mark for $\cos \beta$

$\frac{1}{2}$ mark for $\sin \alpha$

1 mark for substitution into correct identity

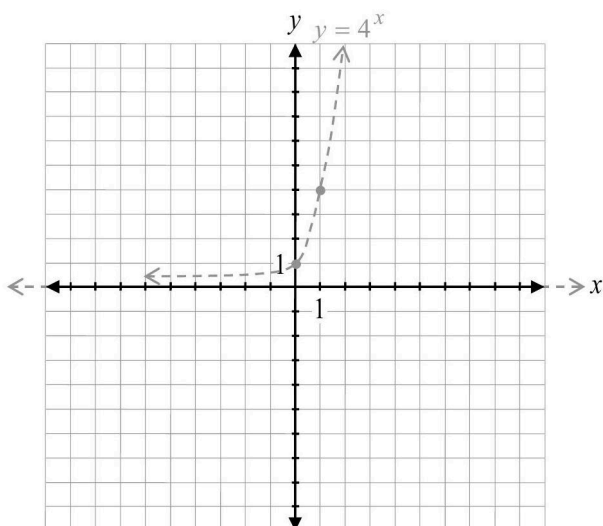
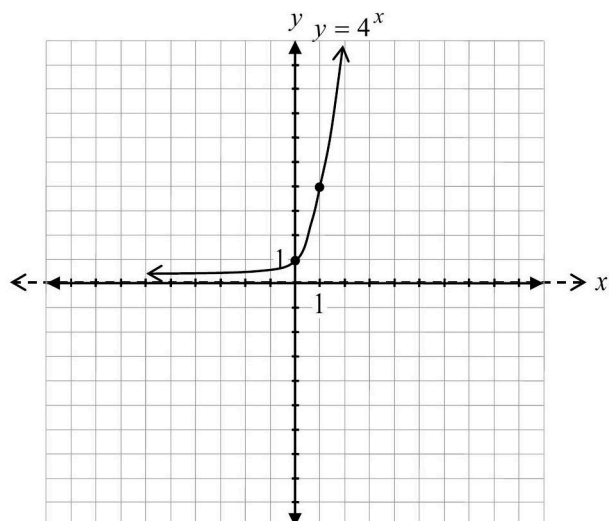
3 marks

Notes:

- Accept any of the following values for x : $x = \pm\sqrt{5}$, $x = \sqrt{5}$, or $x = -\sqrt{5}$.
- Accept any of the following values for y : $y = \pm 12$ or $y = 12$.

Question 32

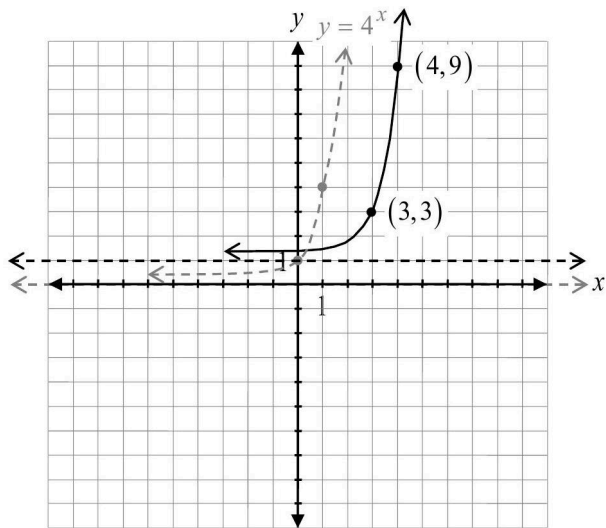
Given the graph of $y = 4^x$, sketch the graph of $y = 2(4)^{x-3} + 1$.



The graph of
 $f(x)$ has already
been drawn for
your reference.
No marks will be
awarded for the
graph of $f(x)$.

Solution

Solution



1 mark for vertical stretch

1 mark for horizontal translation

1 mark for vertical translation

3 marks

Question 33

Determine the coterminal angle of $\frac{\pi}{5}$ over the interval $[-2\pi, 0]$.

Solution

$$-\frac{9\pi}{5}$$

1 mark

Question 34

State the domain of the graph of $y = \log(x - 4) - 8$.

Solution

$$x > 4$$

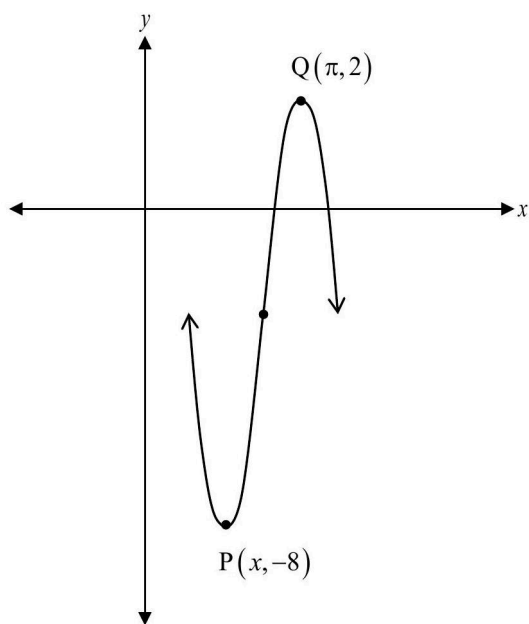
1 mark

or

$$(4, \infty)$$

Question 35

Given the graph of $y = 5 \sin\left[2\left(x + \frac{\pi}{4}\right)\right] - 3$, determine the exact value of the x -coordinate in the point P .



Solution

$$x = \frac{\pi}{2} \quad 1 \text{ mark}$$

Question 36

Verify that the following equation is true for $x = \frac{5\pi}{6}$.

$$\frac{\cos x}{1 - \sin x} = \frac{1 + \sin x}{\cos x}$$

Left-Hand Side	Right-Hand Side

Solution

Left-Hand Side	Right-Hand Side
$\frac{\cos \frac{5\pi}{6}}{1 - \sin \frac{5\pi}{6}}$	$\frac{1 + \sin \frac{5\pi}{6}}{\cos \frac{5\pi}{6}}$
$\frac{-\frac{\sqrt{3}}{2}}{1 - \frac{1}{2}}$	$\frac{1 + \frac{1}{2}}{-\frac{\sqrt{3}}{2}}$ ($\frac{1}{2}$ mark for $\cos \frac{5\pi}{6}$; $1/2$ mark for $\sin \frac{5\pi}{6}$)
$\frac{-\frac{\sqrt{3}}{2}}{\frac{1}{2}}$	$\frac{\frac{3}{2}}{-\frac{\sqrt{3}}{2}}$
$-\frac{\sqrt{3}}{1}$	• $3.\sqrt{3}$ 1 mark for simplification
$-\sqrt{3}$	$-\sqrt{3}$ 2 marks

Question 37

Given that $(x + 1)$ is one of the factors of $P(x) = x^3 - x^2 + kx - 8$, determine the value of k .

Solution

Method 1

$x = -1$	$\frac{1}{2}$ mark for $x = -1$
$0 = (-1)^3 - (-1)^2 + k(-1) - 8$	1 mark for remainder theorem
$0 = -1 - 1 - k - 8$	
$k = -10$	$\frac{1}{2}$ mark for solving for k

Method 2

$x = -1$	$\frac{1}{2}$ mark for $x = -1$

-1	1	-1	k	-8	
		-1	2	$-k - 2$	1 mark for synthetic division
	1	-2	$k + 2$	$-k - 10$	

$-k - 10 = 0$ $k = -10$

$\frac{1}{2}$ mark for solving for k
2 marks

Question 38

Given the function $f(x) = \sqrt{x}$, describe how to use transformations to determine the domain of the function $g(x) = f(x + 2) + 1$.

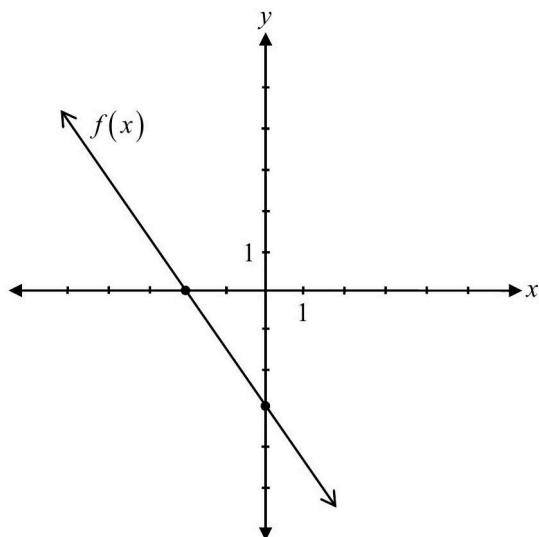
Solution

The graph of $g(x)$ is a horizontal translation 2 units to the left of the graph of $f(x)$, which changes the domain from $x \geq 0$ to $x \geq -2$.

1 mark

Question 39

Given the graph of $y = f(x)$, state the equation of the vertical asymptote of $y = \frac{1}{f(x)}$.



Solution

$$x = -2$$

Question 40

Solve, algebraically.

$$16^x = 64^{2x-1}$$

Solution

$$4^{2x} = 4^{3(2x-1)}$$

$$4^{2x} = 4^{6x-3}$$

$$2x = 6x - 3$$

$$3 = 4x$$

$$\frac{3}{4} = x$$

1 mark for changing to a common base ($\frac{1}{2}$ mark for each)

$\frac{1}{2}$ mark for exponent law

$\frac{1}{2}$ mark for equating exponents

2 marks

Question 41

Given one of the factors of $P(x) = x^3 + 2x^2 - 5x - 6$ is $(x + 3)$, express $P(x)$ in completely factored form.

$P(x) =$

Solution

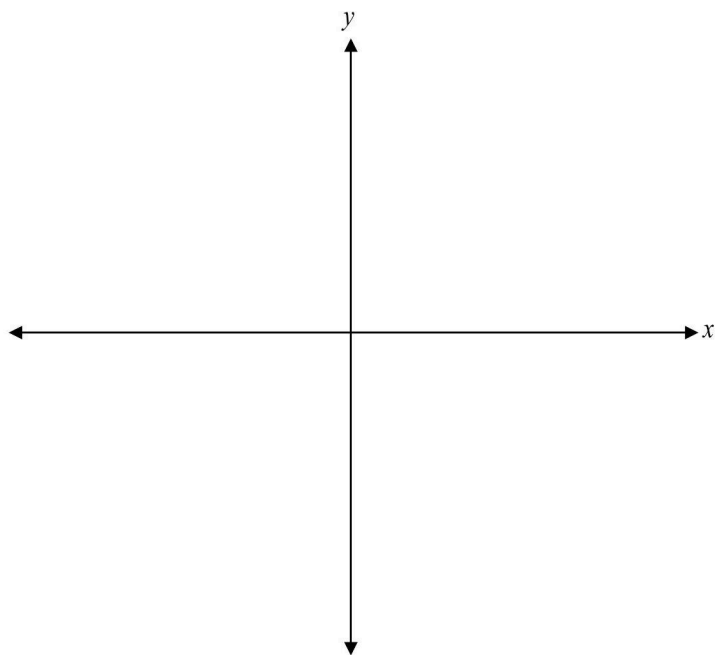
-3	1	2	-5	-6	$\frac{1}{2}$ mark for $x = -3$
	↓	-3	3	6	1 mark for synthetic division (or any equivalent strategy)
	1	-1	-2	0	

$P(x) = (x + 3)(x^2 - x - 2)$
 $P(x) = (x + 3)(x - 2)(x + 1)$ 1/2 mark for consistent product of factors

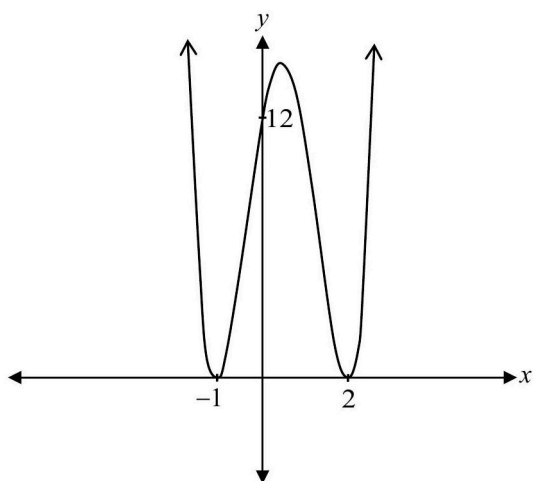
2 marks

Question 42

Sketch the graph of $p(x) = 3(x + 1)^2(x - 2)^2$.



Solution



1 mark for x -intercepts

$\frac{1}{2}$ mark for y -intercept

1 mark for multiplicity of 2 at $x = -1$ and at $x = 2$

($\frac{1}{2}$ mark for each)

$\frac{1}{2}$ mark for end behaviour

3 marks

Question 43

Given that $f(x) = x^2 - 4$ and $g(x) = \sqrt{x}$, determine $f(g(x))$ and state its domain.

$$f(g(x)) =$$

Solution

$$f(g(x)) = (\sqrt{x})^2 - 4$$

$$f(g(x)) = x - 4, x \geq 0$$

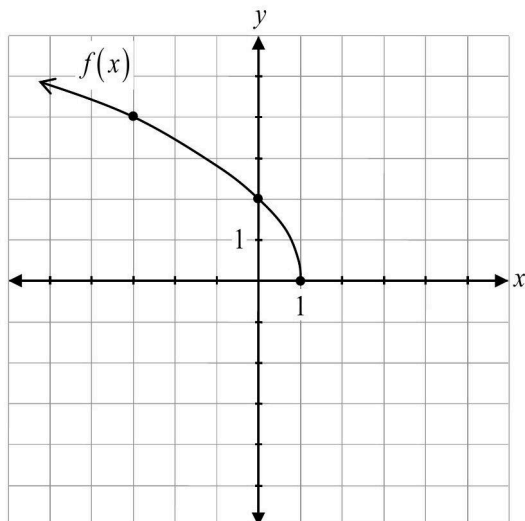
1 mark for composite function

1 mark for domain consistent with composite function

2 marks

Question 44

Determine a possible equation of the function $f(x)$.



$$f(x) =$$

Solution

$$f(x) = 2\sqrt{-(x-1)}$$

or

$$f(x) = \sqrt{-4(x-1)}$$

1 mark for vertical stretch

1 mark for horizontal reflection

1 mark for horizontal translation

3 marks

1 mark for horizontal compression

1 mark for horizontal reflection

1 mark for horizontal translation

3 marks

Question 45

Explain why the graph of $y = \log_2 x$ does not have a y -intercept.

Solution

The domain of the graph is $x > 0$.

1 mark

or

There is a vertical asymptote at $x = 0$.

Question 46

Evaluate.

$$\sin^2\left(-\frac{\pi}{3}\right) + \cos\left(\frac{17\pi}{6}\right) \sec\left(\frac{\pi}{6}\right)$$

Solution

$$\begin{aligned} \left(-\frac{\sqrt{3}}{2}\right)^2 + \left(-\frac{\sqrt{3}}{2}\right)\left(\frac{2}{\sqrt{3}}\right) & \quad 1 \text{ mark for } \sin\left(-\frac{\pi}{3}\right) (1/2 \text{ mark for quadrant; } 1/2 \text{ mark for value}) \\ \frac{3}{4} - 1 & \quad 1 \text{ mark for } \cos\left(\frac{17\pi}{6}\right) (1/2 \text{ mark for quadrant; } 1/2 \text{ mark for value}) \\ -\frac{1}{4} & \quad 1 \text{ mark for } \sec\left(\frac{\pi}{6}\right) (1/2 \text{ mark for quadrant; } 1/2 \text{ mark for value}) \end{aligned}$$

3 marks

Question 47

Determine the coordinates of the point of discontinuity (hole) on the graph of $y = \frac{x^2-3x}{x}$.

Solution

$(0, -3)$ 1 mark for point of discontinuity (hole) at $x = 0$

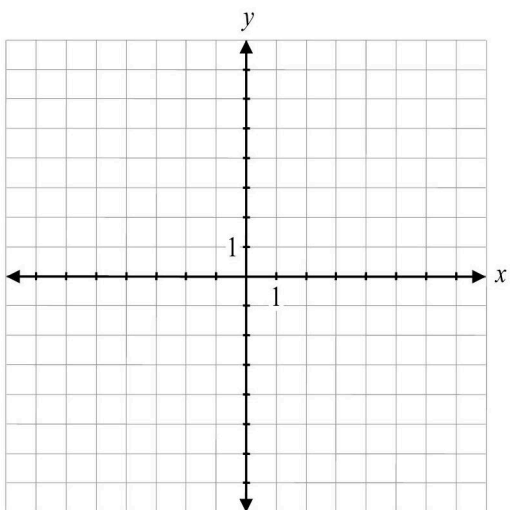
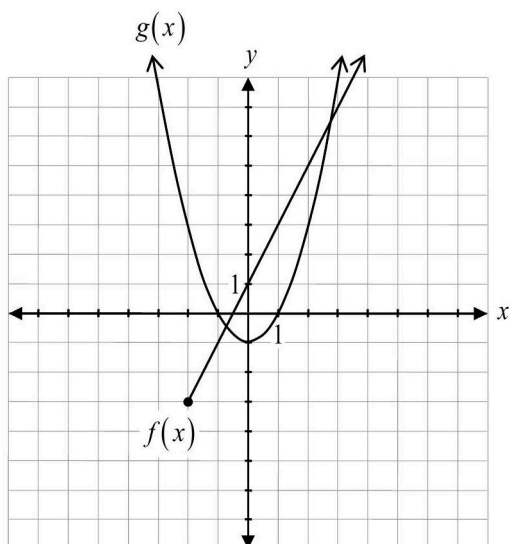
1 mark

Note:

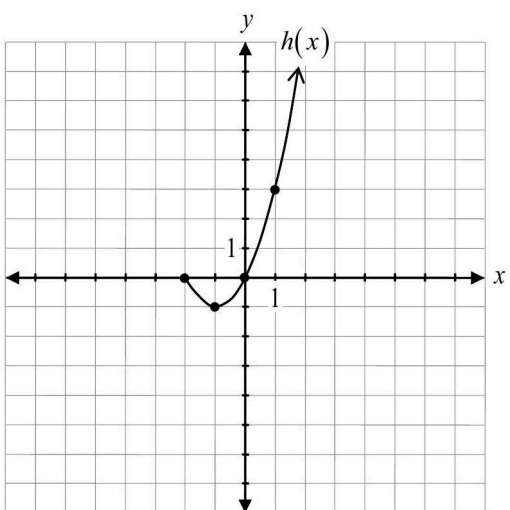
- Deduct $\frac{1}{2}$ mark for procedural error of incorrect y -value.

Question 48

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) + g(x)$.



Solution



1 mark for operation of addition

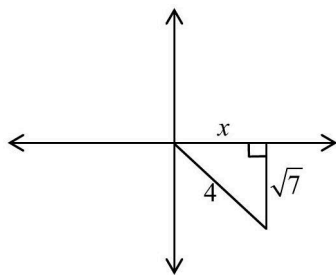
1 mark for restricted domain

2 marks

Question 49

Given that $\csc \theta = -\frac{4}{\sqrt{7}}$ and $\cos \theta > 0$, determine the exact value of $\tan \theta$.

Solution



$$\sin \theta = -\frac{\sqrt{7}}{4}$$

$$x^2 + (\sqrt{7})^2 = (4)^2$$

$$x^2 = 16 - 7$$

$$x^2 = 9$$

$$x = \pm 3$$

$$\tan \theta = -\frac{\sqrt{7}}{3}$$

$\frac{1}{2}$ mark for substitution

$\frac{1}{2}$ mark for solving for x

1 mark for $\tan \theta$ ($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

2 marks

Note:

- Accept any of the following values for x : $x = \pm 3$, $x = 3$, or $x = -3$.